

DDE-SOLVER: A MAPLE PACKAGE FOR DISCRETE DIFFERENTIAL EQUATIONS

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ABSTRACT. We introduce the Maple package **DDE-Solver** for solving Discrete Differential Equations (DDEs) arising in enumerative combinatorics, typically in the enumeration of maps, trees, and walks. These equations are functional equations that algebraically relate a formal power series $F(t, u)$ with specializations of it with respect to the catalytic variable u . In 2006, Bousquet-Mélou and Jehanne proved that when these equations are of a particular form, then F is an algebraic series. This package is based on their work and recent effective algorithms designed by Bostan, Safey El Din, and the author to compute an annihilating polynomial for $F(t, a)$, where a is a constant.

We give a table of running times of the implementation on known examples.

1. INTRODUCTION

Sequences of non-negative integers are ubiquitous in enumerative combinatorics. For instance, consider the set of bicoloured rooted planar maps (with faces either coloured in purple or in white, see Figure 1) such that the degree of each purple face is 3 and the degree of each white face is a multiple of 3. The introduction of the sequence $(c_n)_{n \geq 0}$ of such maps, called *3-constellations* (OEIS A069726), with n purple faces starts with the numbers

$$1, 1, 6, 54, 594, 7371, 99144, \dots$$

For such a sequence, many questions arise: does there exist a recurrence relation satisfied by $(c_n)_{n \geq 0}$? Does there exist a *closed formula* (i.e. an explicit formula) for $(c_n)_{n \geq 0}$? What is the value of c_{n_0} , for $n_0 \in \mathbb{Z}_{\geq 0}$ large (e.g. $n_0 = 3 \cdot 10^6$), and how fast can it be computed? What is the asymptotic growth of c_n when $n \rightarrow \infty$?

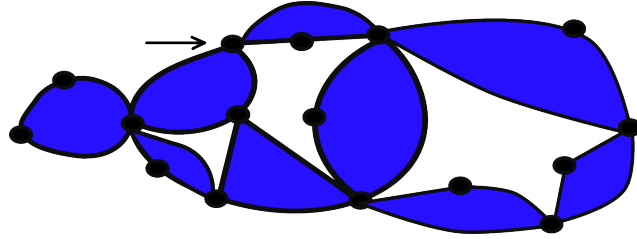


FIGURE 1. A 3-constellation contributing to the coefficient c_6 . Its white internal faces have either degree 3 or 6, and its external white face has degree 6. The arrow indicates the root vertex, and the orientation of the planar map is given by the edge following the root vertex, with the external face on its right.

In order to answer such questions, a common method is to introduce and study the properties of the associated generating function. For the above enumeration, one introduces $G(t) := \sum_{n \geq 0} c_n t^n \in \mathbb{Q}[[t]]$. For sophisticated enumerations, it is usually hard to study directly the generating series. Refining the initial counting problem usually leads to introducing a new variable – called *catalytic* in what follows –, which leads to considering a bivariate generating function, say $F(t, u)$, where u is the catalytic variable. For well-chosen

refinements, it is possible to write an equation over $\mathbb{Q}(t, u)$ relating $F(t, u)$ and $G(t)$.

For the above example of 3-constellations, one typically refines the enumeration with the sequence $(c_{n,d})_{n,d \geq 0}$, where $c_{n,d}$ is the number of 3-constellations having n purple faces and outer degree $3d$. Straightforwardly, one introduces its associated generating function $F(t, u) = \sum_{n,d \geq 0} c_{n,d} u^d t^n \in \mathbb{Q}[u][[t]]$. As $\sum_{d=0}^{\infty} c_{n,d} = c_n$, it holds that $F(t, 1) = G(t)$. After analyzing the construction of these objects (cf. [9, Fig. 7]), one deduces the equation

$$F(t, u) = 1 + tuF(t, u)^3 + tu \left(2F(t, u) + F(t, 1) \right) \frac{F(t, u) - F(t, 1)}{u - 1} + tu \frac{F(t, u) - F(t, 1) - (u - 1) \cdot \partial_u F(t, 1)}{(u - 1)^2}. \quad (1)$$

Now, for $a \in \mathbb{Q}$, denote $\Delta_a : F \in \mathbb{Q}[u][[t]] \mapsto \frac{F(t, u) - F(t, a)}{u - a} \in \mathbb{Q}[u][[t]]$ and Δ_a^ℓ the ℓ^{th} iteration of the *divided difference operator* Δ_a . When $\ell \geq 2$, observe that partial derivatives of F w.r.t. the variable u always appear in $\Delta_a^\ell(F(t, u))$: for instance, when $\ell = 2$, one has

$$\begin{aligned} \Delta_a^2(F(t, u)) &= \Delta_a \left(\frac{F(t, u) - F(t, a)}{u - a} \right), \\ &= \frac{\frac{F(t, u) - F(t, a)}{u - a} - \lim_{u \rightarrow a} \left(\frac{F(t, u) - F(t, a)}{u - a} \right)}{u - a}, \\ &= \frac{\frac{F(t, u) - F(t, a)}{u - a} - \partial_u F(t, a)}{u - a}, \\ &= \frac{F(t, u) - F(t, a) - (u - a) \cdot \partial_u F(t, a)}{(u - a)^2}. \end{aligned}$$

Therefore, (1) is of the form

$$F(t, u) = f(u) + t \cdot Q(F(t, u), \Delta_a F(t, u), \dots, \Delta_a^k F(t, u), t, u), \quad (2)$$

with $k > 0$ and f, Q polynomials (precisely with $k = 2$ and $a = 1$). Equations of the form (2) are called *discrete differential equations (DDEs)* of order k . Observe that one can derive a polynomial recurrence relation on the coefficients in t^n of a solution $F \in \mathbb{Q}[u][[t]]$ of (2), with an initial condition uniquely determined by $F(0, u) = f(u)$. Therefore, (2) admits a unique solution in $\mathbb{Q}[u][[t]]$. Moreover, the following result proved by Bousquet-Mélou and Jehanne in [9], reminiscent of Popescu's theorem [23, Thm. 1.4], implies that this solution is annihilated by a nonzero polynomial with coefficients in $\mathbb{Q}(t, u)$.

Theorem 1. ([9, Thm. 3]) *Let $\mathbb{K}$ be a field of characteristic 0 and consider two polynomials $f \in \mathbb{K}[u]$ and $Q \in \mathbb{K}[x, y_1, \dots, y_k, t, u]$, where $k \in \mathbb{N} \setminus \{0\}$. Let $a \in \mathbb{K}$ and $\Delta_a : \mathbb{K}[u][[t]] \rightarrow \mathbb{K}[u][[t]]$ be the divided difference operator $\Delta_a F(t, u) := (F(t, u) - F(t, a))/(u - a)$. Let us denote by Δ_a^ℓ the operator obtained by iterating ℓ times Δ_a . Then, there exists a unique solution $F \in \mathbb{K}[u][[t]]$ to the functional equation (2), and moreover $F(t, u)$ is algebraic over $\mathbb{K}(t, u)$.*

Applying Theorem 1 to our guiding example, there exists some polynomial $R \in \mathbb{Q}[t, z] \setminus \{0\}$ such that $R(t, G(t)) = 0$. Namely, $G(t) = 1 + t + 6t^2 + 54t^3 + 594t^4 + \dots$ is annihilated by

$$R(t, z) = 81t^2 z^3 - 9t(9t - 2)z^2 + (27t^2 - 66t + 1)z - 3t^2 + 47t - 1.$$

Moreover G is the unique root of R in $\mathbb{Q}[[t]]$. Thus R , together with the initial condition $R(0, z) = 0$, provide a finite amount of data (its monomials) that encode the sequence $(c_n)_{n \geq 0}$. Now, using R , we write the linear differential equation satisfied by $F(t, 1)$

(for instance by using the Maple package [gfun](#) [24] by Salvy and Zimmermann) and deduce (still with [gfun](#)) the associated recurrence relation:

$$(81n^2 + 81n + 18) \cdot c_n - (4n^2 + 14n + 12) \cdot c_{n+1} = 0,$$

with $c_0 = 1, c_1 = 1$. Using this recurrence relation, computing $c_{3 \cdot 10^6}$ only takes 5 seconds in Maple. Solving the recurrence relation, one deduces $c_0 = 1$ and $c_n = \frac{4 \cdot 3^{n-1}}{(2n+2)(2n+1)} \binom{3n}{n}$, for all $n \geq 1$ (OEIS A069726). This latter *closed formula* yields the asymptotic behavior

$$c_n \sim \frac{3^{4n} 2^{-2n}}{2\sqrt{3\pi}} \left(n^{-\frac{5}{2}} + O(n^{-\frac{7}{2}}) \right), \text{ when } n \rightarrow \infty.$$

A central question is to *solve* equations like (2) that is, to compute annihilating polynomials for their solutions. We summarize the objective:

For a solution $F(t, u)$ of a DDE of the form (2), compute $R \in \mathbb{Q}[t, z] \setminus \{0\}$ such that $R(t, F(t, a)) = 0$.

Previous algorithmic works. There exists a rich literature regarding the effective resolution of DDEs. The articles [6, 7, 9] already contain a complete state-of-the-art, we thus only give a quick overview of this literature. Let us start with DDEs such that the polynomial Q in (2) has degree 2 in x . When $k = 1$, Brown introduced in [13] what is now called the *quadratic method*. This method was generalized later by Bender and Canfield in [3] to a particular family of DDEs of arbitrary order, still with Q of degree 2 in x .

Also, the case where Q is linear in $x, y_1, \dots, y_k$ appears in many walks enumeration problems (e.g. [9, §3.1]). Solving DDEs in this case is usually done by applying the *kernel method*, introduced by Knuth in [20] and extended by Bousquet-Mélou and Petkovšek in [10].

Strategies based on the *guess-and-prove* paradigm (popularized in [22]) have been studied by Zeilberger [26] in the case $k = 1$, and improved in this same case with Gessel in [19].

Bousquet-Mélou and Jehanne designed in [9] a general method which can be seen as a generalization of both the kernel and the quadratic methods. Their strategy consists in translating the resolution of a DDE into the resolution of a polynomial system over $\mathbb{Q}(t)$, in $3k$ equations and $3k$ unknowns, with one of its solutions admitting $F(t, a)$ as one of its coordinates. The main step then consists of eliminating all variables except the one that stands for $F(t, a)$. We refer to their algorithm as “duplication”.

Recently, intensive studies have been carried out in [6] (resp. in [7]) in the direction of effectivity to design efficient algorithms for solving DDEs of order $k = 1$ (resp. of any order). The authors introduce three algorithms, referred to as “elimination”, “geometry” and “hybrid”.

The first algorithm, known as the “elimination” algorithm, relies on a geometric observation that allows using only the first 3 equations from the previous paragraph and just $k + 2$ variables. The main step involves leveraging an algebraic-geometric correspondence to interpret the series $F(t, a)$ as the coordinate of a point $\alpha \in \overline{\mathbb{Q}(t)}^k$, obtained by projecting a point of $\overline{\mathbb{Q}(t)}^{k+2}$, where the preimage of α under the projection has a cardinality of at least k .

The second algorithm, known as the “elimination” algorithm, relates to another distinct geometric observation that allows again using only the first 3 equations from the previous paragraph and just $k + 2$ variables. The main step involves leveraging an algebraic-geometric correspondence to interpret the series $F(t, a)$ as the common coordinate of at least k distinct

solutions in $\overline{\mathbb{K}}^3$, where $\mathbb{K}$ is the field obtained from $\mathbb{Q}(t)$ by adjoining all $k + 2$ variables of the 3 equations, except the 3 standing for the series $F(t, u), u, F(t, a)$.

Ultimately, the third algorithm, known as the "hybrid" algorithm, relies on the *guess than prove* paradigm. Namely, it precomputes upper bounds on the partial degrees of the minimal polynomial of $F(t, a)$, then computes a candidate polynomial for annihilating $F(t, a)$, then certify this guess by relying on a *zero lemma*.

Contributions. This paper contains one main contribution. It describes the Maple package [DDE-Solver](#), dedicated to solving DDEs of any order that satisfy the (harmless) assumptions explicited in [7]. Explicitly, [DDE-Solver](#) contains the function

algebraic_single.

This function takes as input a DDE of the form (2), its order k and a set of variables, and outputs a nonzero polynomial annihilating the series $F(t, a)$. Also, some options can be specified, namely: the algorithm that shall be used among "duplication", "elimination", "geometry" and "hybrid"; and an option variable explained in Section 5.

Unlike [7], we have added four new DDEs to the execution time table. These new DDEs include a new family of DDEs, namely m -greedy Tamari lattices [12] (where m is a parameter that goes with the order of the DDEs, and thus increases the difficulty of solving them). In particular, when $m = 3$, this yields the solving of the associated DDE in a few seconds, when the algorithm "duplication" does not terminate in less than a week.

Structure of the paper. In Section 2, we recall why solving a DDE is reduced to eliminating variables in a system of polynomial equations [9, Sec. 2]. Also, we recall in Section 2 different geometric modeling of the underlying problem. In Section 3, we state some preliminaries on effective algebraic geometry that are used in Section 4 when giving an overview on how the algorithms "duplication", "elimination", "geometry" and "hybrid" (resp. [7, §3, §5, §6, §4]) work. In Section 5, we explain how to get started with [DDE-Solver](#), and how to use the options of **algebraic_single**. Finally in Section 6, we provide some timings that illustrate the efficiency of this new package on various DDEs from the literature [4, 8, 9, 12]. In the Appendix (Section 7), we provide explicit Maple code that shows how to apply the various implemented algorithms.

Notations. Denote by $\mathbb{K}$ a field of characteristic 0, by $\overline{\mathbb{K}}$ its algebraic closure, by $\mathbb{Q}$ the field of rational numbers and by $\mathbb{F}_p$ the finite field with p elements. For a fixed positive integer k , denote by $\underline{x}$ (resp. $\underline{u}$ and $\underline{z}$) the variables $x_1, \dots, x_k$ (resp. $u_1, \dots, u_k$ and $z_0, \dots, z_{k-1}$). Denote by $\mathbb{K}[x_1, \dots, x_k]$ the ring of polynomials in the variables $x_1, \dots, x_k$ with coefficients in $\mathbb{K}$. For $P \in \mathbb{K}[\underline{x}]$, denote by $V(P) \subset \overline{\mathbb{K}}^k$ its zero set. For any ideal $\mathcal{I} \subset \mathbb{K}[\underline{x}]$ and any set of polynomials $\mathcal{S} \subset \mathbb{K}[\underline{x}]$, denote by $V(\mathcal{I}) \subset \overline{\mathbb{K}}^k$ (resp. by $V(\mathcal{S}) \subset \overline{\mathbb{K}}^k$) the zero set of $\mathcal{I}$ (resp. of $\mathcal{S}$). Still for $P \in \mathbb{K}[\underline{x}]$, denote by $\text{disc}_{x_j}(P)$ the discriminant [15, Ex. 16, §6, Ch. 3] of P with respect to x_j and by $\text{LC}_{x_j}(P)$ the leading coefficient of P with respect to x_j . Denote by $\overline{\mathbb{K}}[[t^{\frac{1}{d}}]]$ the ring $\bigcup_{d \geq 1} \overline{\mathbb{K}}[[t^{\frac{1}{d}}]]$ of Puiseux series in t with positive fractional exponents.

2. FROM DDEs TO POLYNOMIAL SYSTEMS

Starting from a DDE of the form (2), we multiply it by the least power of $(u - a)$ such that we obtain a polynomial functional equation of the form

$$P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, \quad (3)$$

for some polynomial $P \in \mathbb{Q}[x, z_0, \dots, z_{k-1}, t, u]$. Taking the derivative of (3) with respect to u yields by the chain rule

$$\begin{aligned} \partial_u F(t, u) &\cdot \partial_x P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) \\ &+ \partial_u P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0. \end{aligned} \quad (4)$$

Consider the equation in u given by

$$\partial_x P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0. \quad (5)$$

Observe that the solutions of (5) that belong to $\overline{\mathbb{Q}}[[t^{\frac{1}{*}}]]$ are also solutions, by using (4), of the equation in u given by

$$\partial_u P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0.$$

Thus any non constant solution in $\overline{\mathbb{Q}}[[t^{\frac{1}{*}}]]$ of (5) is a solution in u of the system of constraints

$$\begin{cases} P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, \\ \partial_x P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, & u(u - a) \neq 0, \\ \partial_u P(F(t, u), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0. \end{cases} \quad (6)$$

The below hypothesis needs to be satisfied in order to make the ongoing general strategy work. This assumption holds for the examples considered in this paper.

Global assumption: There exist k distinct solutions $U_1(t), \dots, U_k(t) \in \overline{\mathbb{Q}}[[t^{\frac{1}{*}}]]$ (GA) to the system of equations (6). Also, $U_1(t), \dots, U_k(t) \notin \overline{\mathbb{Q}}$.

In [7], the authors studied three geometric interpretations of (GA) that we recall below. The first one is introduced in [9], while the last two were introduced respectively in [7, §5, §6]. Also, it is a nontrivial consequence of the proof of [9, Thm. 3] that the series $\{\partial_u^{i-1} F(t, a), U_i(t), F(t, U_i(t))\}_{1 \leq i \leq k}$ considered in this paper are elements of $\overline{\mathbb{Q}}(t)$.

Duplication approach. This approach was introduced by Bousquet-Mélou and Jehanne in [9, Sec. 2] and works as follows. It results from (GA) that the following relations hold:

$$\forall 1 \leq i \leq k, \begin{cases} P(F(t, U_i(t)), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, U_i(t)) = 0, \\ \partial_x P(F(t, U_i(t)), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, U_i(t)) = 0, \\ \partial_u P(F(t, U_i(t)), F(t, a), \dots, \partial_u^{k-1} F(t, a), t, U_i(t)) = 0, \end{cases}$$

and

$$\prod_{1 \leq i < j \leq k} (U_i(t) - U_j(t)) \cdot \prod_{1 \leq i \leq k} U_i(t) \cdot (U_i(t) - a) \neq 0.$$

In terms of polynomial equations, the above relations are equivalent to saying that the *duplicated* polynomial system $\mathcal{S}_{\text{dup}}$ defined by

$$\forall 1 \leq i \leq k, \begin{cases} P(x_i, z_0, \dots, z_{k-1}, t, u_i) = 0, \\ \partial_x P(x_i, z_0, \dots, z_{k-1}, t, u_i) = 0, \\ \partial_u P(x_i, z_0, \dots, z_{k-1}, t, u_i) = 0, \end{cases} \quad m \cdot \prod_{1 \leq i < j \leq k} (u_i - u_j) \cdot \prod_{1 \leq i \leq k} u_i \cdot (u_i - a) - 1 = 0,$$

admits the nontrivial solutions

$$x_i = F(t, U_i(t)), \quad u_i = U_i(t), \quad z_{i-1} = \partial_u^{i-1} F(t, a), \quad \text{for } 1 \leq i \leq k. \quad (7)$$

Note that (7) uniquely determines the value associated to the variable m . Also, the polynomial system $\mathcal{S}_{\text{dup}}$ admits $3k + 1$ equations and unknowns (t is a parameter as we work over $\mathbb{Q}(t)$); that is, the polynomials in $\mathcal{S}_{\text{dup}}$ are seen as elements of the ring $\mathbb{Q}(t)[m, \underline{x}, \underline{z}, \underline{u}]$. Denote by $V(\mathcal{S}_{\text{dup}})$ the solution set of $\mathcal{S}_{\text{dup}}$ in $\overline{\mathbb{Q}(t)}^{3k+1}$.

Elimination theory approach. This second approach was introduced in [7, Sec. 5]. It follows from (6) and (GA) that the system of polynomial constraints defined in $\overline{\mathbb{Q}(t)}[x, u]$ by:

$$\begin{cases} P(x, F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, \\ \partial_x P(x, F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, & u(u - a) \neq 0, \\ \partial_u P(x, F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, \end{cases} \quad (8)$$

admits the k solutions $(x, u) = (F(t, U_i(t)), U_i(t)) \in \overline{\mathbb{Q}(t)}^2$, for $1 \leq i \leq k$. Note that these solutions have distinct u -coordinates. This observation can be reformulated geometrically in the following way. Consider the geometric projection $\pi : (x, u, \underline{z}) \in \overline{\mathbb{Q}(t)}^{k+2} \mapsto (\underline{z}) \in \overline{\mathbb{Q}(t)}^k$, and denote by $\mathcal{X} \subset \overline{\mathbb{Q}(t)}^{k+2}$ the solution set of the polynomial constraints

$$\begin{cases} P(x, z_0, \dots, z_{k-1}, t, u) = 0, \\ \partial_x P(x, z_0, \dots, z_{k-1}, t, u) = 0, & u(u - a) \neq 0, \\ \partial_u P(x, z_0, \dots, z_{k-1}, t, u) = 0. \end{cases} \quad (9)$$

For $\alpha \in \overline{\mathbb{Q}(t)}^k$, denote by $\#_u(\mathcal{X}, \alpha)$ the number of u -coordinates in $\pi^{-1}(\alpha) \cap \mathcal{X}$ that are not constants and define $\mathcal{F}_k(u, \mathcal{X}) := \{\alpha \in \overline{\mathbb{Q}(t)}^k \mid \#_u(\mathcal{X}, \alpha) \geq k\}$. Under assumption (GA), it holds that $(F(t, a), \dots, \partial_u^{k-1} F(t, a)) \in \mathcal{F}_k(u, \mathcal{X})$.

Geometric approach. This approach was introduced in [7, Sec. 6]. In addition to having (6), it follows from (GA) that the system of polynomial constraints defined in $\overline{\mathbb{Q}(t)}[x, z_1, u]$ by

$$\begin{cases} P(x, F(t, a), z_1, \partial_u^2 F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, \\ \partial_x P(x, F(t, a), z_1, \partial_u^2 F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, & u(u - a) \neq 0, \\ \partial_u P(x, F(t, a), z_1, \partial_u^2 F(t, a), \dots, \partial_u^{k-1} F(t, a), t, u) = 0, \end{cases} \quad (10)$$

admits the k solutions $(x, z_1, u) = (F(t, U_i(t)), \partial_u F(t, a), U_i(t)) \in \overline{\mathbb{Q}(t)}^3$, for $1 \leq i \leq k$. Note that these solutions are distinct. Denote by $\tilde{z}_1$ the set of variables $z_0, z_2, \dots, z_{k-1}$. Consider the geometric projection $\pi_{\tilde{z}_1} : (x, u, \underline{z}) \in \overline{\mathbb{Q}(t)}^{k+2} \mapsto (\tilde{z}_1) \in \overline{\mathbb{Q}(t)}^{k-1}$ and define

$$\mathcal{S}_k(\mathcal{X}) := \{\alpha = (\alpha_0, \dots, \alpha_{k-1}) \in \overline{\mathbb{Q}(t)}^k \mid \alpha \in \pi(\mathcal{X}) \wedge \# \mathcal{X} \cap \pi_{\tilde{z}_1}^{-1}((\alpha_0, \alpha_2, \dots, \alpha_{k-1})) \geq k\}.$$

Under assumption (GA), it holds that $(F(t, a), \dots, \partial_u^{k-1} F(t, a)) \in \mathcal{S}_k(\mathcal{X})$.

Summary. In Section 2, we defined for each of the three approaches a solution set which contains a point whose z_0 -coordinate is the series $F(t, a)$. The goal of Section 4 is to recall how the algorithms studied in [7, §3, §5, §6] compute a polynomial characterization of these sets. We then deduce from this characterization an annihilating polynomial of $F(t, a)$.

3. SOME PRELIMINARIES ON POLYNOMIAL TOOLS

The current section is devoted to the introduction of the tools used in Section 4.

3.1. ELIMINATION THEORY THROUGH GRÖBNER BASES

For $x_1, \dots, x_n$ some variables and $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{Z}_{\geq 0}^n$, denote $\mathbf{x}^\alpha := x_1^{\alpha_1} \cdots x_n^{\alpha_n}$. Also for $S \subset \mathbb{K}[x_1, \dots, x_n]$, denote by $\langle S \rangle$ the ideal generated by S in $\mathbb{K}[x_1, \dots, x_n]$.

Monomial orders and Gröbner bases. Elimination monomial orderings attribute to a variable (or to a block of variables) which shall be eliminated a larger weight than the weights of the other variables. An example is the lexicographic order $\succ_{\text{lex}}$.

Definition 1. ([15, Def. 3, §2, Ch. 2]) Let $\alpha = (\alpha_1, \dots, \alpha_n)$ and $\beta = (\beta_1, \dots, \beta_n)$ be in $\mathbb{Z}_{\geq 0}^n$. We say that $\mathbf{x}^\alpha \succ_{\text{lex}} \mathbf{x}^\beta$ if the leftmost nonzero entry of the vector $\alpha - \beta \in \mathbb{Z}^n$ is positive.

Example 1. In $\mathbb{K}[x_1, x_2]$: $x_1^4 x_2^2 \succ_{\text{lex}} x_1^3 x_2^{10}$, $x_1^2 x_2 \succ_{\text{lex}} x_1$, $x_1^2 \succ_{\text{lex}} x_2^5$ and $x_1^3 x_2^2 > x_1^3 x_2$.

Another useful family of monomial orders that we will refer to in Section 4 is the family of graded monomial orders. An example is the graded reverse lexicographic order $\succ_{\text{grevlex}}$.

Definition 2. ([15, Def. 6, §2, Ch. 2]) Let $\alpha = (\alpha_1, \dots, \alpha_n)$ and $\beta = (\beta_1, \dots, \beta_n)$ be in $\mathbb{Z}_{\geq 0}^n$. We say that $\mathbf{x}^\alpha \succ_{\text{grevlex}} \mathbf{x}^\beta$ if $|\alpha| = \sum_{i=1}^n \alpha_i > |\beta| = \sum_{i=1}^n \beta_i$, or if $|\alpha| = |\beta|$ and the rightmost nonzero entry of $\alpha - \beta \in \mathbb{Z}^n$ is negative.

Example 2. In $\mathbb{K}[x_1, x_2, x_3]$: $x_1^5 x_2^7 x_3 \succ_{\text{grevlex}} x_1^4 x_2^2 x_3^3$ and $x_1 x_2^4 x_3^2 \succ_{\text{grevlex}} x_1^3 x_2 x_3^3$.

We invite readers unfamiliar with monomial orders (see [15, Def. 1, §2, Ch. 2] for a general definition) to choose one of the two above monomial orders $\succ_{\text{lex}}$ or $\succ_{\text{grevlex}}$ for the next definition. Also, for $p \in \mathbb{K}[x_1, \dots, x_n]$ a polynomial and $\succ$ a monomial order on $\mathbb{K}[x_1, \dots, x_n]$, denote by $\text{LT}_\succ(p)$ the leading term of p with respect to the monomial order $\succ$.

Definition 3. ([15, Def. 5, §5, Ch. 2]) Fix a monomial order $\succ$ on the ring $\mathbb{K}[x_1, \dots, x_n]$. A finite subset $G = \{g_1, \dots, g_s\}$ of an ideal $\mathcal{I} \subset \mathbb{K}[x_1, \dots, x_n]$ different from $\{0\}$ is said to be a Gröbner basis of $\mathcal{I}$ for the order $\succ$ if $\langle \text{LT}(g_1), \dots, \text{LT}(g_n) \rangle = \langle \text{LT}(g) \mid g \in \mathcal{I} \rangle$.

A fundamental property ([15, Cor. 6, §5, Ch. 2]) of Gröbner bases is, with the notations of Definition 3, that such a basis G always exists, and that G generates $\mathcal{I}$, that is $\langle G \rangle = \mathcal{I}$.

Gröbner bases and projections. The results below justify the use of Gröbner bases as a dedicated theoretical and computational tool, allowing one to characterize projections as solution sets of conjunctions of polynomial equations and inequations.

Theorem 2 (Elimination theorem). ([15, Thm. 2, §1, Ch. 3]) Let $\mathcal{I} \subset \mathbb{K}[x_1, \dots, x_n]$ be an ideal. Denote by G a Gröbner basis of $\mathcal{I}$ with respect to the order $\succ_{\text{lex}}$. Then for every $0 \leq \ell \leq n-1$, the set $G_\ell = G \cap \mathbb{K}[x_{\ell+1}, \dots, x_n]$ is a Gröbner basis of the ideal $\mathcal{I}_\ell := \mathcal{I} \cap \mathbb{K}[x_{\ell+1}, \dots, x_n]$.

By Theorem 2, the set of polynomials G_ℓ finitely generates $\mathcal{I}_\ell$. Now, denote the geometric projection $\pi_\ell : (x_1, \dots, x_n) \in \mathbb{K}^n \mapsto (x_{\ell+1}, \dots, x_n) \in \mathbb{K}^{n-\ell}$. Recall that a Zariski closed set in $\mathbb{K}^n$ is defined as the solution set of some polynomial equations defined in $\mathbb{K}[x_1, \dots, x_n]$.

Theorem 3 (Closure theorem). ([15, Thm. 3, §2, Ch. 3]) Let $\mathcal{I} \subset \mathbb{K}[x_1, \dots, x_n]$ be an ideal and $V(\mathcal{I}) \subset \mathbb{K}^n$ be its zero set. Consider $\mathcal{I}_\ell := \mathcal{I} \cap \mathbb{K}[x_{\ell+1}, \dots, x_n]$. Then $V(\mathcal{I}_\ell)$ is the smallest Zariski closed set containing $\pi_\ell(V(\mathcal{I})) \subset \mathbb{K}^{n-\ell}$.

Theorems 2 and 3 imply that characterizing the Zariski closure of the projection of a zero set $V(\mathcal{I})$ onto some coordinate subspace is done by computing a Gröbner basis for $\succ_{\text{lex}}$.

Theorem 4 (Extension theorem). ([15, Thm. 2, §5, Ch. 3]) Let $G = \{g_1, \dots, g_s\}$ be a Gröbner basis of $\mathcal{I} \subset \mathbb{K}[x_1, \dots, x_n]$ for the order $\succ_{\text{lex}}$. For each $1 \leq j \leq s$, consider

$$g_j = c_j(x_2, \dots, x_n) \cdot x_1^{N_j} + (\text{terms in which } x_1 \text{ has degree} < N_j),$$

where $N_j \geq 0$ and $x_j \in \mathbb{K}[x_2, \dots, x_n]$ is nonzero. Assume $\alpha = (\alpha_2, \dots, \alpha_n) \in V(\mathcal{I} \cap \mathbb{K}[x_2, \dots, x_n])$ is a partial solution with the property that $\alpha \notin V(c_1, \dots, c_s)$. Then

$$\{f(x_1, \alpha) \mid f \in \mathcal{I}\} = \{g_{j_0}(x_1, \alpha)\},$$

where $g_{j_0} \in G$ satisfies $c_{j_0}(\alpha) \neq 0$ and g_{j_0} has minimal degree in x_1 among all elements g_j with $c_j(\alpha) \neq 0$. Furthermore, if $g_{j_0}(\alpha_1, \alpha) = 0$ for $\alpha_1 \in \overline{\mathbb{K}}$, then $(\alpha_1, \alpha) \in V(\mathcal{I})$.

Theorem 4 implies that characterizing projections (and not their Zariski closure) is done by considering the disjunction of inequations associated to the non simultaneous vanishing of the $c_1, \dots, c_s$ (notations of Theorem 4): for each condition in this disjunction, one shall add the vanishing of all the elements of G_ℓ of Theorem 2.

3.2. COUNTING SPECIFIC SOLUTIONS

Let n, s be positive integers. Consider the notation $\mathbf{x} = x_1, \dots, x_n$ and $\mathbf{y} = y_1, \dots, y_s$. Let $\mathcal{I} \subset \mathbb{K}[\mathbf{x}]$ be a radical ideal such that its zero set $V(\mathcal{I}) \subset \overline{\mathbb{K}}^n$ is finite. For $\alpha \in V(\mathcal{I})$, denote by $[x_1](\alpha)$ the x_1 -coordinate of α . Denote the projection $\pi_{x_1} : (\mathbf{x}) \in \overline{\mathbb{K}}^n \mapsto x_1 \in \overline{\mathbb{K}}$. Let $g \in \mathbb{K}[\mathbf{y}][z]$ and denote $\text{LC}_z(g) \in \mathbb{K}[\mathbf{y}]$ the leading coefficients of g in z .

Problem 5. Let ℓ be a positive integer. Characterize with polynomial equations defined in $\mathbb{K}[\mathbf{x}]$ the set $\{\beta \in V(\mathcal{I}) \mid \pi_{x_1}^{-1}([x_1](\beta)) \cap V(\mathcal{I}) \geq \ell\}$.

Problem 6. Let $1 \leq \ell \leq \deg_z(g)$. Characterize with polynomial inequations defined in $\mathbb{K}[\mathbf{y}]$ the points $\beta \in \overline{\mathbb{K}}^s \setminus V(\text{LC}_x(g))$ such that $g(\mathbf{y} = \beta, z)$ has at least ℓ distinct roots.

Answer to Problem 5. Denote $\mathcal{A} := \mathbb{K}[\mathbf{x}]/\mathcal{I}$. It results from the finiteness of $V(\mathcal{I})$ that $\mathcal{A}$ has finite dimension as a $\mathbb{K}$ -vector space ([15, Thm.6, §3, Ch.5]). Define $m_{x_1} : f \in \mathcal{A} \mapsto x_1 \cdot f \in \mathcal{A}$ to be the multiplication map by x_1 in $\mathcal{A}$, and denote by $\chi_{x_1} \in \mathbb{K}[T]$ its characteristic polynomial. The following results from the radicality of $\mathcal{I}$ and from [16, Prop. 2.7, §2, Ch. 4].

Fact 1. We have the equality $\chi_{x_1} = \prod_{\alpha \in V(\mathcal{I})} (T - [x_1](\alpha))$.

As a consequence of Fact 1, we obtain the following set equality:

$$\begin{aligned} \{\beta \in V(\mathcal{I}) \mid \#\pi_{x_1}^{-1}([x_1](\beta)) \cap V(\mathcal{I}) \geq \ell\} \\ = \{\beta \in \overline{\mathbb{K}}^n \mid \beta \in V(\mathcal{I}) \wedge \chi_{x_1}([x_1](\beta)) = 0 \wedge \dots \wedge \partial_T^{\ell-1} \chi_{x_1}([x_1](\beta)) = 0\}. \end{aligned} \quad (11)$$

Equality (11) provides a conjunction of polynomial equations that answers Problem 5.

Answer to Problem 6. Define the Hermite quadratic form associated with g by

$$H_g : (f, h) \in (\mathbb{K}(\mathbf{y})[z]/\langle g \rangle)^2 \mapsto \text{Trace}(m_{f \cdot h}) \in \mathbb{K}(\mathbf{y}),$$

where $\text{Trace}(\cdot)$ is the trace operator and $m_{f \cdot h}$ is the multiplication map by $f \cdot h$ in $\mathbb{K}(\mathbf{y})[z]/\langle g \rangle$. Also, denote by $M_{H_g} \in \mathbb{K}(\mathbf{y})^{\deg_z(g) \times \deg_z(g)}$ the matrix of H_g in the basis $\{1, z, \dots, z^{\deg_z(g)-1}\}$.

The following fact is an immediate consequence of [16, Thm.5.2, §5, Ch.2] and of the observation that, by performing euclidian divisions by g in $\mathbb{K}(\mathbf{y})[z]$, the denominator of the image of z^i in $\mathbb{K}(\mathbf{y})[z]/\langle g \rangle$ can only be a power of $\text{LC}_z(g)$.

Fact 2. *The denominators in the matrix M_{H_g} are powers of $\text{LC}_z(g)$. Moreover if $1 \leq \ell \leq \deg_z(g)$, then the points $\beta \in \overline{\mathbb{K}}^s \setminus V(\text{LC}_z(g))$ at which $g(\mathbf{y} = \beta, z)$ admits less than ℓ distinct solutions are the points $\beta \in \overline{\mathbb{K}}^s \setminus V(\text{LC}_z(g))$ at which the $\ell \times \ell$ -minors of M_{H_g} all vanish.*

Fact 2 implies that the conjunction of polynomial inequations answering Problem 6 is given by the non vanishing of $\text{LC}_z(g)$ and by the non vanishing of the $\ell \times \ell$ -minors of M_{H_g} .

3.3. CHANGE OF MONOMIAL ORDERING

For n generic elements of $\mathbb{K}[x_1, \dots, x_n]$ of degree d , computing a Gröbner basis of $\mathcal{I}$ for the order $\succ_{\text{grevlex}}$ has an arithmetic cost which is in $d^{O(n)}$ [2, Prop.1].

Let $\mathcal{I} \subset \mathbb{K}[x_1, \dots, x_n]$ be an ideal whose solution set is finite. The following fact uses the FGLM ([17]) algorithm which, on input a Gröbner basis of $\mathcal{I}$ for the order $\succ_{\text{grevlex}}$, outputs a Gröbner basis of $\mathcal{I}$ for the order $\succ_{\text{lex}}$. The complexity of this algorithm is, under some genericity assumptions, in $\tilde{O}(D^\omega)^1$ (see [21]), where D is the cardinality of $V(\mathcal{I})$ ($\mathcal{I}$ is assumed radical). By the Bézout bound, and with the notations of the above paragraph, we have the bound $D \leq d^n$, this makes the following fact useful for computing in $d^{O(n)}$ a Gröbner bases of $\mathcal{I}$ for a $\succ_{\text{lex}}$ order.

Fact 3. *On input an ideal $\mathcal{I} \subset \mathbb{K}[x_1, \dots, x_n]$ whose solution set $V(\mathcal{I}) \subset \overline{\mathbb{K}}^n$ is finite, there exists an algorithm that computes a Gröbner basis $G_{\succ_{\text{grevlex}}}$ of $\mathcal{I}$ for the order $\succ_{\text{grevlex}}$ and that turns it into a Gröbner basis $G_{\succ_{\text{lex}}}$ of $\mathcal{I}$ for the order $\succ_{\text{lex}}$.*

With the notations as above and without using the algorithm FGLM [17] mentioned in Fact 3, the computation of a Gröbner basis of $\mathcal{I}$ for the order $\succ_{\text{lex}}$ has an arithmetic cost bounded by $C_2 d^{C_3 n^3}$ [14], for $C_2, C_3 \in \mathbb{Z}_{\geq 0}$.

4. FUNCTION IMPLEMENTED IN DDE-SOLVER

The function implemented in **DDE-Solver** is:

algebraic_single

In this section, we explain the algorithms on which this function, and its options, rely.

The function **algebraic_single** computes a non trivial annihilating polynomial of the series $F(t, a)$. Four algorithms can be applied: “duplication”, “elimination”, “geometry”, “hybrid” (resp. studied in [7, §3, §5, §6, §4]). As the algorithm “hybrid” relies on the three other algorithms, we first describe “duplication”, “elimination” and “geometry”.

In Section 2, we defined solution sets that contain a point whose z_0 -coordinate is $F(t, a)$:

- “duplication”: $V(\mathcal{S}_{\text{dup}})$ contains the point defined by (7),
- “elimination”: $\mathcal{F}_k(u, \mathcal{X})$ contains the point $(F(t, a), \dots, \partial_u^{k-1} F(t, a))$,
- “geometric”: $\mathcal{S}_k(\mathcal{X})$ contains the point $(F(t, a), \dots, \partial_u^{k-1} F(t, a))$.

¹We denote by ω the constant of multiplication matrix that is, multiplying two matrices in $\mathbb{K}^{n \times n}$ can be done by using $\tilde{O}(n^\omega)$ arithmetic operations in $\mathbb{K}$, where $\tilde{O}(\cdot)$ is a $O(\cdot)$ that hides polylogarithmic factors.

We introduce a finiteness assumption that holds for the examples studied in this paper.

Finiteness assumption: The sets $V(\mathcal{S}_{\text{dup}})$, $\mathcal{F}_k(u, \mathcal{X})$ and $\mathcal{S}_k(\mathcal{X})$ are finite. (FA)

General strategy. Under (FA), the general spirit of the algorithms from [7, §3, §5, §6] is:

- (i) To compute a disjunction of polynomial equations whose solution set is the set of interest,
- (ii) To eliminate all variables except (z_0, t) from the polynomial characterization (i).

Let us apply the above steps to the three approaches from Section 2.

4.1. OPTION “DUPLICATION”

Step (i) of the general strategy: The polynomial system $\mathcal{S}_{\text{dup}}$ characterizes $V(\mathcal{S}_{\text{dup}})$.

Step (ii) of the general strategy: Denote by $\mathcal{I}_{\text{dup}} \subset \mathbb{Q}(t)[m, x, z, u]$ the ideal generated by the polynomials in $\mathcal{S}_{\text{dup}}$. By assumption (FA), it is possible to apply to $\mathcal{I}_{\text{dup}}$ the algorithm from Fact 3. This algorithm outputs, for a proper choice of variable ordering, a Gröbner basis $G_{\succ_{\text{lex}}}$ of $\mathcal{I}_{\text{dup}}$ which contains an element of $\mathbb{Q}(t)[z_0]$. Applying Theorem 3 to the projection onto the z_0 -coordinate space together with Theorem 2 yields, under (FA), that this polynomial is nonzero. Denote by $R \in \mathbb{Q}[t, z_0]$ its numerator. By [7, Prop. 2], we have $R(t, F(t, a)) = 0$.

Annex 7.1 illustrates the solving of (1) via this method.

4.2. OPTION “ELIMINATION”

Recall from Section 2 that $\mathcal{F}_k(u, \mathcal{X}) = \{\alpha \in \overline{\mathbb{Q}(t)}^k \mid \#_u(\mathcal{X}, \alpha) \geq k\}$. The algorithm from [7, §5] requires the additional assumption that for any $\alpha \in \overline{\mathbb{Q}(t)}^k$, $\#_u(\mathcal{X}, \alpha) < +\infty$. Under this assumption and (FA), the algorithm designed in [7, §5] works as follows.

Step (i) of the general strategy:

- (1) Compute using Theorems 2, 4 successive disjunctions of conjunctions of polynomial equations and inequations defined in $\mathbb{K}(t)[u, \underline{z}]$ whose solution sets are successively the projections onto the $(u, \underline{z})$ -coordinate space and onto the $(\underline{z})$ -coordinate space. The union of the polynomial constraints in these disjunctions have the form $\{I, E\}$, where I is a set of inequations defined in $\mathbb{Q}(t)[u, \underline{z}]$ and E is a set of equations defined in $\mathbb{Q}(t)[u, \underline{z}]$. Without loss of generality, assume that the polynomials in E are a Gröbner basis of the ideal they generate, for $\succ_{\text{lex}}$ with u greater than $\underline{z}$.
- (2) Apply Theorem 4 to E : at $\underline{z} = \alpha \in \overline{\mathbb{Q}(t)}^k$ fixed, the cardinality condition in the definition of $\mathcal{F}_k(u, \mathcal{X})$ is equivalent to studying for a polynomial g_{j_0} in E :
 - if g_{j_0} has its leading coefficient in u that does not vanish at $\underline{z} = \alpha$ and, by Theorem 4, if g_{j_0} is the polynomial in E of minimal degree in u that satisfies this property: checking this minimality condition yields a disjunction of polynomial equations and inequations defined in $\mathbb{Q}(t)[\underline{z}]$,
 - for this g_{j_0} , it remains to add a conjunction of polynomial conditions in $\mathbb{Q}(t)[u, \underline{z}]$ so that $g_{j_0}(u, \underline{z} = \alpha)$ has at least k distinct solutions:
If $\deg_u(g_{j_0}) < k$, adds the vanishing of all the coefficients (in u) of g_{j_0} to E ; else adds the conjunction of inequations given by the non vanishing of the $k \times k$ -minors of $M_{H_{g_{j_0}}}$ from Fact 2.

- (3) Steps 1 and 2 above computed a disjunction of polynomial equations and inequations defined in $\mathbb{Q}(t)[u, z]$ whose solution set in $\overline{\mathbb{Q}(t)}^{k+1}$ is $\mathcal{F}_k(u, \mathcal{X})$. As by (FA) this set is finite, this polynomial characterization can be turned into a disjunction of conjunction of polynomial equations by using Rabinowitsch trick² in order to remove the solution set given by the inequations: the introduced variables are eliminated using Theorem 2. Denote by $\mathcal{D}_{\text{elim}}$ the so-obtained disjunction.

Step (ii) of the general strategy: We perform this step in the same spirit as what we did in Section 4.1. By (FA), the solution set of $\mathcal{D}_{\text{elim}}$ is finite. We can thus apply the algorithm from Fact 3 to each disjunction in $\mathcal{D}_{\text{elim}}$. Each application of this algorithm allows to compute a nonzero polynomial of $\mathbb{Q}(t)[z_0]$: this polynomial might be equal to 1 when there is no solution to the studied conjunction. We consider the numerators of all these polynomials and denote by $R \in \mathbb{Q}[t, z_0]$ their product. By [7, Prop. 5.5], we have $R(t, F(t, a)) = 0$.

Annex 7.2 illustrates the solving of (1) via this method.

4.3. OPTION “GEOMETRY”

Recall from Section 2 that

$$\mathcal{S}_k(\mathcal{X}) := \{\alpha = (\alpha_0, \dots, \alpha_{k-1}) \in \overline{\mathbb{Q}(t)}^k \mid \alpha \in \pi(\mathcal{X}) \wedge \# \mathcal{X} \cap \pi_{z_1}^{-1}((\alpha_0, \alpha_2, \dots, \alpha_{k-1})) \geq k\}.$$

The algorithm from [7, §6] requires many additional technical assumptions (see the beginning of [7, §6]). We do not make them explicit for sake of simplicity. In the rest of this section, denote by $\succ_{\text{bgrevlex}}$ the block monomial order over $\mathbb{Q}(t)[m, x, u, z]$ defined as follows:

- We use the monomial order $\succ_{\text{grevlex}}$ on each of the two blocks $\{m, x, u, z_1\}$ and $\{z_1\}$,
- Two monomials are compared w.r.t. the variables m, x, u, z_1 . In case of equality, they are compared w.r.t. the variables z_1 .

Example 3. For $k = 3$:

$$x^3 u z_1^2 m \succ_{\text{bgrevlex}} z_0^{20} z_2^3, \quad x^3 u z_1^2 m z_2 \succ_{\text{bgrevlex}} x^3 u z_1^2 m z_0 \quad \text{and} \quad z_0^4 \succ_{\text{bgrevlex}} z_0^2 z_2.$$

Under these assumptions and (FA), the algorithm designed in [7, §6] works as follows.

Step (i) of the general strategy:

- (1) Compute a Gröbner basis G of the ideal $\mathcal{I} := \langle P, \partial_x P, \partial_u P, m \cdot u(u - a) - 1 \rangle \subset \mathbb{Q}(t)[m, x, u, z]$ for the order $\succ_{\text{bgrevlex}}$.
- (2) Compute, using normal form computations modulo $\mathcal{I}$ ³, the matrix M_{z_1} of the multiplication map m_{z_1} in the quotient ring $\mathbb{Q}(t, z_1)[m, x, u, z_1]/j(\mathcal{I})$, where $j(\mathcal{I})$ is the image of the injective map $j : \mathbb{Q}(t)[m, x, u, z] \rightarrow \mathbb{Q}(t, z_1)[m, x, u, z_1]$.
- (3) Compute the characteristic polynomial $\chi_{z_1} \in \mathbb{Q}(t, z_1)[T]$. Using the assumptions of [7, §6], we can replace χ_{z_1} by its numerator with respect to the variables z_1 , so that we now have $\chi_{z_1} \in \mathbb{Q}(t)[z_1, T]$.
- (4) Denote by $\mathcal{D}_{\text{geom}}$ the set of equations given by the vanishing of the polynomials in G and of the new conditions $(\chi_{z_1})|_{T=z_1} = 0 \wedge \dots \wedge (\partial_T^{k-1} \chi_{z_1})|_{T=z_1} = 0$.

Step (ii) of the general strategy: We perform this step in the same spirit as we did in Section 4.1. By (FA), the solution set of $\mathcal{D}_{\text{geom}}$ is finite. We thus apply the algorithm from Fact 3

²Instead of considering an inequation $Q \neq 0$ (for a given polynomial $Q \in \mathbb{K}[x_1, \dots, x_n]$), introduce an extra variable m and consider the equation $m \cdot Q - 1 = 0$ defined in $\mathbb{K}[m, x_1, \dots, x_n]$.

³Normal form computations modulo ideals are generalizations of euclidean divisions by polynomials.

to the ideal generated by the polynomials associated with the equations in $\mathcal{D}_{\text{geom}}$. This algorithm outputs a nonzero polynomial of $\mathbb{Q}(t)[z_0]$. Consider the numerator of this polynomial and denote it by $R \in \mathbb{Q}[t, z_0]$. By [7, Prop. 6.4], we have $R(t, F(t, a)) = 0$.

Annex 7.3 illustrates the solving of (1) via this method.

4.4. OPTION “HYBRID”

The *hybrid guess-and-prove* algorithm [7, §4] works as follows:

- (1) Compute bounds b_{z_0}, b_t such that $\deg_t(R) \leq b_t$ and $\deg_{z_0}(R) \leq b_{z_0}$, for some nonzero polynomial $R \in \mathbb{Q}[t, z_0]$ annihilating $F(t, a)$ (we detail this step in the next section),
- (2) Compute the truncated series $F(t, a) \bmod t^{2b_t b_{z_0} + 1}$,
- (3) Guess a polynomial $M \in \mathbb{Q}[t, z_0]$ such that $M(t, F(t, a)) = O(t^{(b_t+1)(b_{z_0}+1)-1})$,
- (4) Check that $M(t, F(t, a)) = O(t^{2b_t b_{z_0} + 1})$.

When step 4 is satisfied, applying [7, Prop. 5] yields $M(t, F(t, a)) = 0$.

Annex 7.4 illustrates the solving of (1) via this method.

4.5. DETAILS OF IMPLEMENTATIONS

As we target practical efficiency, the implementations of the options “duplication”, “elimination” and “geometry” incorporate the following key improvements from computer algebra.

To reduce the computations of $\mathbb{Q}(t)$ to $\mathbb{Q}$, we perform evaluation-interpolation on t .

We rely on fast multi-modular arithmetic. This consists in applying multiple times a given algorithm with the base field $\mathbb{Q}$ replaced by successive distinct prime fields $\mathbb{F}_p$. These computations output various images mod p of the same $R \in \mathbb{Q}[t, z_0]$. From there, one lifts the modular coefficients over $\mathbb{Q}$ by applying the Chinese Remainder Theorem (CRT) together with rational numbers reconstruction [18, §5.10].

Implementation of the option “hybrid”. Computing the bounds (b_t, b_{z_0}) in step 1 is done using the option “elimination” of Section 4.2. More precisely, computing b_{z_0} is done by specializing t at a random value θ of some random prime field $\mathbb{F}_p$ and performing the option “elimination” with base field $\mathbb{F}_p$: it outputs $R(\theta, z_0)$ and yields $b_{z_0} = \deg_{z_0}(R(\theta, z_0))$ (a similar computation is done for computing b_t). The generation of terms is done using a divide and conquer approach which computes the first terms of $F(t, u) \bmod t^{2b_t b_{z_0} + 1}$ before specializing them to $u = a$. Finally, the guessing is done by computing Hermite-Padé approximants. For this, we use the function `seriestoalgeq` of the Maple package `gfun` [24].

5. GETTING STARTED

5.1. INSTALLATION AND CUSTOMIZATION

Files provided: `DDE-Solver` is available in the “.mla” format. If one wishes to customize the package, the Maple scripts “`ddesolver.mpl`” and “`build.mpl`” can be downloaded on the dedicated [github webpage](#). Used as follows, they allow to modify the package by modifying “`ddesolver.mpl`” and to generate the corresponding version of “`ddesolver.mla`” by executing “`build.mpl`”.

Customization of DDE-Solver: Replacing “PATH/TO” in the Maple file “build.mpl” by the relevant path to the file where “ddesolver.mpl” and “build.mpl” are located allows one, by executing “build.mpl”, to generate a new version of “ddesolver.mla”.

Loading DDE-Solver: The Maple variable `libname` shall be set so that “ddesolver.mla” is located in a visible place.

```
libname := "/home/yourname/ddesolver/lib", libname:
```

Once `libname` has been correctly set up, one executes in Maple

```
with(ddesolver);
```

in order to load and use the package.

5.2. USING DDE-SOLVER

Input/Output syntax. The arguments of `algebraic_single` are (P, k, var) , where $P \in \mathbb{Q}[x, z_0, \dots, z_{k-1}, t, u]$ is the polynomial in (3), k is the order of the considered DDE and `var` is $[x, z_0, \dots, z_{k-1}, t, u]$ (precisely in that order).

The output of `algebraic_single` (P, k, var) is some nonzero polynomial $R \in \mathbb{Q}[t, z_0]$ such that $R(t, F(t, a)) = 0$.

Example 4. Consider (1). Multiplying (1) by $(u - 1)^2$ yields

$$0 = (u - 1)^2(1 - F(t, u) + tuF(t, u)^3) + tu(u - 1)(2F(t, u) + F(t, 1))(F(t, u) - F(t, 1)) \\ + tu(F(t, u) - F(t, 1) - (u - 1)\partial_u F(t, 1)).$$

Thus $P := (u - 1)^2(1 - x + tux^3) + tu(u - 1)(2x + z_0)(x - z_0) + tu(x - z_0 - (u - 1)z_1)$ and $k := 2$. Continuing the analysis with Maple

```
P := (u-1)^2(1-x+tux^3)+tu(u-1)(2x+z0)(x-z0)+tu(x-z0-(u-1)z1);
with(ddesolver):
annihilating_polynomial(P, 2, [x, z0, z1, t, u]);
```

outputs

$$(16tz_0^2 - 8tz_0 + t - 16) * \\ (81t^2z_0^3 - 81t^2z_0^2 + 27t^2z_0 + 18tz_0^2 - 3t^2 - 66tz_0 + 47t + z_0 - 1)$$

Thus $R := (16tz_0^2 - 8tz_0 + t - 16)(81t^2z_0^3 - 81t^2z_0^2 + 27t^2z_0 + 18tz_0^2 - 3t^2 - 66tz_0 + 47t + z_0 - 1)$ is an annihilating polynomial of $F(t, 1)$. A direct analysis on the solutions of $R(t, z_0)$ that are finite at $t = 0$ shows that the second factor of R is the minimal polynomial of $F(t, 1)$.

Options. It is possible to benefit in practice from two options: the choice of the *algorithm*, and the choice of the *variable* (either t or z_0) on which we perform evaluation–interpolation.

Choice of the algorithm. Four algorithms are implemented: “duplication”, “elimination”, “geometry” (only when $k = 2$) and “hybrid”. By default, the algorithm used is “elimination”.

Choice of the variable. The two choices on which we perform the evaluation–interpolation strategy are t and z_0 . The default choice is t . This option affects the timings (see Figure 2). Both choices yield the same output.

The choice of the algorithm and of the variable on which we perform the evaluations is made by executing

algebraic_single($P, k, \text{var}, \text{algorithm}, \text{variable}$).

Note that these two options must be either not specified at all, or specified in the same call.

Example 5. Continuing Example 4 with $\text{algorithm} = \text{"geometry"}$ and $\text{variable} = z_0$ yields

`annihilating_polynomial(P, 2, [x, z0, z1, t, u], 'geometry', z0);`

(16tz0²-8tz0+t-16)

(81t²z0³-81t²z0²+27t²z0+18tz0²-3t²-66tz0+47t+z0-1)

We refer to the next section for a discussion on the choice of the parameter variable.

6. EXAMPLES

6.1. IMPACT OF THE OPTION variable

The motivation of this study is the following. Assume that we perform evaluation–interpolation on z_0 . The chosen algorithm (e.g. “elimination” from Section 4.1) will be run a first time with t specialized (in order to get the degree in z_0 of the output $R \in \mathbb{Q}[t, z_0]$), then it will be run multiple times at z_0 specialized (in order to evaluate–interpolate R w.r.t z_0). Hence if the algorithm is faster at t specialized than at z_0 specialized (and if the partial degrees of the output R are “close”), then the choice of the evaluation–interpolation variable has a significant impact on the timings. This is what happens below.

Example 6. Consider the DDE of the enumeration of 3-Tamari lattices [8, Prop. 8]:

$$F(t, u) = u + tuF(t, u) \frac{F(t, u) \frac{F(t, u) - F(t, 1)}{u-1} - F(t, 1) \partial_u F(t, 1)}{u-1} - \frac{F(t, 1)^2 \partial_u^2 F(t, 1)}{2} - F(t, 1) \partial_u F(t, 1)^2, \quad (12)$$

Multiplying (12) by $(u-1)^3$ yields the functional equation $P(F(t, u), F(t, 1), \partial_u F(t, 1), t, u) = 0$, where $P := -tu^3xz_0^2z_2 - 2tu^3xz_0z_1^2 - 2tu^2x^2z_0z_1 + 2tu^2xz_0^2z_2 + 4tu^2xz_0z_1^2 + 2tux^4 - 2tux^3z_0 + 2tux^2z_0z_1 - tuxz_0^2z_2 - 2tuxz_0z_1^2 + 2u^4 - 2u^3x - 6u^3 + 6u^2x + 6u^2 - 6ux - 2u + 2x \in \mathbb{Q}[x, z_0, z_1, z_2, t, u]$. We thus have in Maple

```
P := -tu^3xz0^2z2-2tu^3xz0z1^2-2tu^2x^2z0z1+2tu^2xz0^2z2+4tu^2xz0z1^2
      +2tux^4-2tux^3z0+2tux^2z0z1-tuxz0^2z2-2tuxz0z1^2
      +2u^4-2u^3x-6u^3+6u^2x+6u^2-6ux-2u+2x;
```

```
with(ddesolver):
```

```
time(annihilating_polynomial(P, 2, [x, z0, z1, z2, t, u]));
      # variable = t (implicitly chosen)
```

2089.599

```
time(annihilating_polynomial(P, 2, [x, z0, z1, z2, t, u],
      'elimination', z0)); # variable = z0
141.897
```

```
R := annihilating_polynomial(P, 2, [x, z0, z1, z2, t, u],
      'elimination', z0);
```

$$\begin{aligned}
R := & t^5 z_0^{16} + 135 t^4 z_0^{13} + 1024 t^4 z_0^{12} + 7290 t^3 z_0^{10} - 1762560 t^3 z_0^9 \\
& + 393216 t^3 z_0^8 + 196830 t^2 z_0^7 + 111694464 t^2 z_0^6 + 580976640 t^2 z_0^5 \\
& + t(67108864 t + 2657205) z_0^4 - 661978656 t z_0^3 + 4721836032 t z_0^2 \\
& + (-8371830784 t + 14348907) z_0 + 4294967296 t - 14348907
\end{aligned}$$

6.2. PRACTICAL RESULTS OF THE FUNCTION `ALGEBRAIC_SINGLE`

Figure 2 gathers some timings obtained after using the function

`algebraic_single`.

Each column is associated with a DDE taken from the literature ([4, 8, 9, 12]). For each DDE, we precise: its order k , the variable on which is performed the evaluation–interpolation, the algorithm used and the bi-degree of the output $R \in \mathbb{Q}[t, z_0]$.

From Table 1, we strengthen the message of Section 6.1 and draw some conclusions:

- *Choice of the algorithm.* The best algorithm to choose depends in an important way on the DDE. Even among DDEs of the same family (e.g. DDEs [4], [5] and [8]) the answer is not clear: the first two are solved faster with the option `algorithm = "elimination"` while the last one is solved faster with `algorithm = "hybrid"`. The efficiency of this last option `algorithm = "hybrid"` comes from the fact that the output polynomial has small bidegree (3, 9), thus allowing a fast computation of the first terms of $F(t, 1)$; and at the same time, that the computations are hard with the other algorithms.

This impossibility to predict which algorithm is more efficient for an input DDE justifies that we let the user choose the algorithm which shall be applied.

- *Choice of the variable.* As in Section 6.1, the table shows that the choice of the variable (z_0 or t) on which we perform evaluation–interpolation is important. Let us pick one example to illustrate it. For the DDE [3], performing evaluation–interpolation over t with `algorithm = "geometry"` takes 2 minutes while it takes 54 minutes when performing evaluation–interpolation over z_0 . With the notations of Section 4.3, the reason is that at z_0 specialized to some random $\theta \in \mathbb{F}_p$, the main computational cost (in addition to the Gröbner basis computation) is to compute the discriminant w.r.t. T of $(\chi_{z_1})|_{z_0=\theta} \in \mathbb{Q}[t, T]$, whose partial degrees are 6 in T and 423 in t . In the same time at t specialized to some random $\nu \in \mathbb{F}_p$, the polynomial $(\chi_{z_1})|_{t=\nu} \in \mathbb{Q}[z_0, T]$ has degree 15 in z_0 and 6 in T : it yields a faster computation of its discriminant w.r.t. T .

As all these intermediate computational data are hard to predict from the input DDE, we let the user choose on which variable the evaluation–interpolation shall be performed.

- *Solving DDEs previously out of reach!* The DDEs [7], [8] are solved via theoretical arguments ([11, 12]). Here, it is the first time that they are solved in a “click”.

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Data	[1]		[2]		[3]		[4]		[5]		[6]		[7]		[8]	
k	2		2		2		2		3		3		3		4	
eval–interp variable	z_0	t	z_0	t	z_0	t	z_0	t	z_0	t	z_0	t	z_0	t	z_0	t
“duplication”	0.5s	0.7s	1.6s	2.3s	41m*	10m*	2.7s	1.5s	2m45s	2m50s	13h	27h	∞	∞	∞	∞
“elimination”	0.2s	0.3s	0.4s	0.4s	1h20m*	4m*	1.7s	1.2s	1m10s	45s	2m20s	35m	1m	7m30s	2d19h	2d
“geometry”	2s	2.2s	1.6s	0.9s	54m*	2m*	0.8s	0.7s	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$
“hybrid”	50s		2m		∞		1.4s		18s		1h42m		34s		2h41m	
$(\deg_t(R), \deg_{z_0}(R))$	(4, 9)		(2, 3)		(132, 6)		(3, 5)		(3, 7)		(5, 16)		(2, 4)		(3, 9)	

FIGURE 2. Practical results of **algebraic_single** on DDEs from the literature.

Computations are done in Maple on a computer equipped with Intel® Xeon® Gold CPU 6246R v4 @ 3.40GHz and 1.5TB of RAM with 1 thread. Gröbner bases computations are performed using the C library [msolve](#) [5], implemented by Berthomieu, Eder and Safey El Din.

- [1]: Enumeration of 2-Tamari lattices, [8, Prop. 8] for $m = 3$,
- [2]: Enumeration of 2-greedy Tamari intervals, [12, Prop. 3.1] for $m = 2$,
- [3]: Enumeration of non-separable near-triangulations in which all intern vertices have degree at least 5, [4, Prop. 4.3, Eq. 22],
- [4]: Enumeration of 3-constellations, [9, Equation 29],
- [5]: Enumeration of 4-constellations, [9, Prop. 12] for $m = 4$,
- [6]: Enumeration of 3-Tamari lattices, [8, Prop. 8] for $m = 4$,
- [7]: Enumeration of 3-greedy Tamari intervals, [12, Prop. 3.1] for $m = 3$,
- [8]: Enumeration of 5-constellations. [9, Prop. 12] for $m = 5$.

- s, m, h, d: seconds, minutes, hours, days.
- ∞ : The computations did not finish within 5 days,
- $\times$: The algorithm is not implemented for $k > 2$,
- *: We added the condition $m' \cdot (1 + t^3)(1 - t^3)t - 1 = 0$ to the input polynomial constraints.

7. APPENDIX

7.1. SOLVING 3-CONSTELLATIONS USING SECTION 4.1

The DDE [9, Eq. 29] associated with the enumeration of 3-constellations is given by

$$F(t, u) = 1 + tuF(t, u)^3 + tu \left(2F(t, u) + F(t, 1) \right) \frac{F(t, u) - F(t, 1)}{u - 1} + tu \frac{F(t, u) - F(t, 1) - (u - 1)\partial_u F(t, 1)}{(u - 1)^2} \quad (13)$$

Multiplying (13) by $(u - 1)^2$ yields $P(F(t, u), F(t, 1), \partial_u F(t, 1), t, u) = 0$, where

$$P := (u - 1)^2(1 - x + tx^3) + tu(u - 1)(2x + z_0)(x - z_0) + tu(x - z_0 - (u - 1)z_1) \in \mathbb{Q}[x, z_0, z_1, t, u].$$

Define the polynomial system

$$\begin{aligned} \mathcal{S}_{\text{dup}} := & \left(P(x_1, z_0, z_1, t, u_1) = 0, \partial_x P(x_1, z_0, z_1, t, u_1) = 0, \partial_u P(x_1, z_0, z_1, t, u_1) = 0, \right. \\ & P(x_2, z_0, z_1, t, u_2) = 0, \partial_x P(x_2, z_0, z_1, t, u_2) = 0, \partial_u P(x_2, z_0, z_1, t, u_2) = 0, \\ & \left. m \cdot (u_1 - u_2) \cdot (u_1 - 1) \cdot (u_2 - 1) \cdot u_1 \cdot u_2 - 1 = 0 \right), \end{aligned}$$

and let $\mathcal{I}_{\text{dup}} \subset \mathbb{Q}(t)[m, x_1, x_2, z_0, z_1, u_1, u_2]$ be the ideal generated by the polynomials in $\mathcal{S}_{\text{dup}}$.

The algorithm described in Section 4.1:

- (1) Computes a generator $R \in \mathcal{I}_{\text{dup}} \cap \mathbb{Q}(t)[z_0]$ by applying Fact 3 and finds

$$R = (81t^2z_0^3 - 81t^2z_0^2 + 27t^2z_0 + 18tz_0^2 - 3t^2 - 66tz_0 + 47t + z_0 - 1)(16tz_0^2 - 8tz_0 + t - 16).$$

7.2. SOLVING 3-CONSTELLATIONS USING SECTION 4.2

Define $\mathcal{S} := (P, \partial_x P, \partial_u P, m \cdot u(u - 1) - 1)$, for P as in Appendix 7.1. Denote by $\mathcal{X} \subset \overline{\mathbb{Q}(t)}^4$ the solution set of the conditions $P = 0 \wedge \partial_x P = 0 \wedge \partial_u P = 0 \wedge u(u - 1) \neq 0$ in $\overline{\mathbb{Q}(t)}^4$.

Goal: Characterize $\mathcal{F}_k(u, \mathcal{X}) := \{\alpha \in \overline{\mathbb{Q}(t)}^2 \mid \#_u(\mathcal{X}, \alpha) \geq 2\}$ as the solution set of some explicit polynomial constraints and deduce a nonzero $R \in \mathbb{Q}[t, z_0]$ such that

$$R(t, F(t, 1)) = 0.$$

The algorithm described in Section 4.2:

- (1) Characterizes the projection of $\mathcal{X}$ onto the (u, z_0, z_1) -coordinate space:

```
G := Groebner[Basis](S, lexdeg([m, x], [u, z0, z1, t])):
E := Groebner[Basis](remove(has, G, {m, x}), lexdeg([u], [z0, z1, t])):
Leading := select(has, G, {m, x}):
```

The first element of Leading has the form

$$1534(u - 1) \cdot x + (\text{polynomial in } u, z_0, z_1, t).$$

As $\mathcal{X} \cap \{u = 1\} = \emptyset$, it results from the extension theorem that the zero set of E in $\overline{\mathbb{Q}(t)}^3$ is precisely the projection of $\mathcal{X}$ onto the (u, z_0, z_1) -coordinate space.

- (2) Identifies algebraic conditions in E describing the cardinality of the fiber in $\mathcal{F}_k(u, \mathcal{X})$:

```

# Dealing with the polynomials of degree < 2 in u in E:
NewConditions := [];
for i to nops(E) do
  if degree(E[i], u) < 2 then
    NewConditions := [op(NewConditions), coeffs(E[i], u)]:
  fi:
od:
E := Groebner[Basis]([op(E), op(NewConditions)],
  lexdeg([u], [z0, z1, t]]):
# Dealing with the polynomials of degree > 1 in u in E:
map(p -> degree(p, u), E);
# prints the degrees in u of the polynomials in E
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3]

```

In order to have $\#_u(\mathcal{X}, \alpha) \geq 2$ for a given $\alpha \in \overline{\mathbb{Q}(t)}^2$, we shall exclude the solution set of $\text{disc}_u(E[10])$:

```
sat := discrim(E[10], u):
```

Denote by $c_{10}(z_0, z_1) \in \mathbb{Q}(t)[z_0, z_1]$ the coefficients of u^2 in $E[10]$. As $V(E) \cap \{c_{10} = 0\} = \emptyset$, the algorithm deduces from Theorem 4 that $\mathcal{F}_k(u, \mathcal{X})$ is the solution set of the polynomial equations given by E and of the inequation $\text{sat} \neq 0$.

- (3) Turns the inequation $\text{sat} \neq 0$ into the equation $m \cdot \text{sat} - 1 = 0$ by introducing an extra variable m . Finally, it deduces a nonzero polynomial $R \in \mathbb{Q}[t, z_0]$ annihilating $F(t, 1)$.

```

H := Groebner[Basis]([op(E), m*sat-1], lexdeg([m, u, z1], [z0, t]]):
op(remove(has, H, {m, u, z1}));

```

$$(16tz_0^2 - 8tz_0 + t - 16)(81t^2z_0^3 - 81t^2z_0^2 + 27t^2z_0 + 18tz_0^2 - 3t^2 - 66tz_0 + 47t + z_0 - 1)$$

Thus

$$R = (16tz_0^2 - 8tz_0 + t - 16)(81t^2z_0^3 - 81t^2z_0^2 + 27t^2z_0 + 18tz_0^2 - 3t^2 - 66tz_0 + 47t + z_0 - 1).$$

7.3. SOLVING 3-CONSTELLATIONS USING SECTION 4.3

Denote by $P \in \mathbb{Q}[x, z_0, z_1, t, u]$ the polynomial in Appendix 7.1 and by $\mathcal{S}$ the set of polynomials $(P, \partial_x P, \partial_u P, m \cdot tu(u-1) - 1) \subset \mathbb{Q}(t)[m, x, z_0, z_1, u]$. Moreover, recall that $\mathcal{X} \subset \overline{\mathbb{Q}(t)}^4$ is the solution set of the constraints $P = 0 \wedge \partial_x P = 0 \wedge \partial_u P = 0 \wedge u(u-1) \neq 0$ and that

$$\mathcal{S}_k(\mathcal{X}) := \{\alpha = (\alpha_0, \dots, \alpha_{k-1}) \in \overline{\mathbb{Q}(t)}^k \mid \alpha \in \pi(\mathcal{X}) \wedge \# \mathcal{X} \cap \pi_{z_1}^{-1}((\alpha_0, \alpha_2, \dots, \alpha_{k-1})) \geq k\}.$$

Goal: Characterize $\mathcal{S}_k(\mathcal{X})$ as the solution set of some polynomial constraints and deduce a nonzero $R \in \mathbb{Q}[t, z_0]$ such that $R(t, F(t, 1)) = 0$.

The algorithm described in Section 4.3:

- (1) Computes a Gröbner basis of the ideal generated by $\mathcal{S}$ for the order $\succ_{\text{bgrelex}}$:

```
G := Groebner[Basis](S, lexdeg([m, x, u, z1], [t, z0]));
```

- (2) Computes the matrix M_{z_1} of the multiplication map $m_{z_1} : f \mapsto z_1 \cdot f$ in the quotient ring $\mathbb{Q}(t, z_0)[m, x, u, z_1]/\langle j(\mathcal{S}) \rangle$, where j is the usual inclusion map of $\mathbb{Q}(t)[m, x, z_0, z_1]$ in $\mathbb{Q}(t, z_0)[m, x, u, z_1]$:

```
L_1, L_2 := Groebner[NormalSet](subs(t = rand(), z0 = rand(), G),
                                tdeg(m, x, u, z1)):
M := Groebner[MultiplicationMatrix](z1, L1, L2, G, tdeg(m, x, u, z1)):
```

- (3) Computes the characteristic polynomial of M_{z_1} and defines χ_{z_1} to be its numerator:

```
xi_z1 := LinearAlgebra[CharacteristicPolynomial](M, T):
chi_z1 := factor( numer(xi_z1) ):
```

- (4) Thus $S_k(\mathcal{X})$ is the solution set of the constraints $P = 0 \wedge \partial_x P = 0 \wedge \partial_u P = 0 \wedge u(u-1) \neq 0 \wedge \text{disc}_T(\chi_{z_1}) = 0$. An annihilating polynomial of $F(t, 1)$ is given by the discriminant $R \in \mathbb{Q}[t, z_0]$ of χ_{z_1} with respect to T , that is

$$R = 12230590464t^{14} \cdot (16tz_0^2 - 8tz_0 + t - 16)^2 \cdot (1 + (3z_0 - 1)^2t^2 + (-6z_0 - 14)t)^3 \cdot (tz_0 + 1)^{12} \cdot (81t^2z_0^3 - 9t(9t - 2)z_0^2 + (27t^2 - 66t + 1)z_0 - 3t^2 + 47t - 1)^2.$$

7.4. SOLVING 3-CONSTELLATIONS USING SECTION 4.4

Consider (13) and the polynomial $P \in \mathbb{Q}[x, z_0, z_1, t, u]$ of Appendix 7.1. Define the set of polynomials $\mathcal{S} := (P, \partial_x P, \partial_u P, m \cdot u(u-1) - 1) \in \mathbb{Q}(t)[m, x, z_0, z_1, u]$.

Goal: Compute a nonzero polynomial $R \in \mathbb{Q}[t, z_0]$ such that $R(t, F(t, 1)) = 0$.

The algorithm described in Section 4.4:

- (1) Applies twice the strategy described in 7.2 to P . It draws at random a prime number p and some $\theta \in \mathbb{F}_p$, say $p = 12301$ and $\theta = 1328$, and applies 7.2 with $\mathcal{S}$ replaced by $(\mathcal{S})|_{t=\theta}$ and $\mathbb{Q}$ replaced by $\mathbb{F}_p$. It outputs

$$7957(z_0 + 11829)(z_0^3 + 4863z_0^2 + 8711z_0 + 3012)(z_0 + 6622)$$

The degree being 5, it sets $b_{z_0} = 5$. Similarly, it obtains $b_t = 3$.

- (2) Computes the 32 first terms of $F(t, 1)$:

$$F_1 = 1 + t + 6t^2 + 54t^3 + 594t^4 + \cdots + 913075994651156584840651326232625946t^{31} \bmod t^{32}$$

- (3) Uses the function `seriestoalgeq` of the Maple package `gfun` [24]:

```
collect(subs(T(t) = z0,
             gfun[seriestoalgeq](series(F1, t, degree(F1, t)), T(t))) [1],
        z0, factor);
```

$$81t^2z_0^3 - 9t(9t - 2)z_0^2 + (27t^2 - 66t + 1)z_0 - 3t^2 + 47t - 1$$

The above polynomial, denoted by M , is a candidate for annihilating $F(t, 1)$.

- (4) Proves the guessed polynomial with the below lines

```
series(subs(z0 = F1, M), t, 31);
```

$$0(t^{31})$$

$$\text{Thus } R = 81t^2z_0^3 - 9t(9t - 2)z_0^2 + (27t^2 - 66t + 1)z_0 - 3t^2 + 47t - 1.$$

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